

As a reminder the matrix are :

$${}^0T_1 = \begin{pmatrix} c_{\theta_1} & c_{\theta_1} & c_{\theta_1} & p_1 \\ c_{\theta_1} & s_{\theta_1} & 0 & 0 \\ s_{\theta_1} & c_{\theta_1} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} {}^1T_2 &= c_{\theta_2} c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_2} s_{\theta_3} c_{\theta_4} - s_{\theta_2} s_{\theta_3} c_{\theta_5} - s_{\theta_2} s_{\theta_3} c_{\theta_4} c_{\theta_5} \\ {}^1T_3 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} - s_{\theta_2} s_{\theta_3} s_{\theta_4} \\ {}^1T_4 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_5 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_6 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_7 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_8 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_9 &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{10} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{11} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{12} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{13} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{14} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{15} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{16} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{17} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{18} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{19} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \\ {}^1T_{20} &= c_{\theta_2} (-c_{\theta_3} c_{\theta_4} c_{\theta_5} - s_{\theta_3} s_{\theta_4} c_{\theta_5}) - s_{\theta_2} s_{\theta_3} c_{\theta_4} - c_{\theta_2} s_{\theta_3} s_{\theta_4} c_{\theta_5} \end{aligned}$$

So following Z-Y-X Euler angles we have

$$\begin{aligned} \beta &= \arctan2(-r_{3,1}, \sqrt{r_{1,1}^2 + r_{2,1}^2}) \\ \alpha &= \arctan2(r_{1,1}, r_{2,1}) \\ \gamma &= \arctan2(r_{3,1}, r_{3,2}) \end{aligned}$$

There is a singularity if $|r_{3,1}| = 1$

In this case we take $\gamma = 0$ and $\alpha = \arctan2(r_{1,1}, r_{2,1})$

And the inverse kinematic gave us:

$$\begin{aligned} \Delta < 0 & \text{ no solution } (c_2 \in [-1, 1]) \\ \Delta = 0 & \quad \theta_2 = \pm \arccos\left(\frac{-B}{2A}\right) \\ \Delta > 0 & \quad \theta_2 = \pm \arccos\left(\frac{-B \pm \sqrt{\Delta}}{2A}\right) \end{aligned}$$

$$\theta_4 = \arctan2\left(\pm \sqrt{1 - \left(\frac{-d_1 - c_2 d_2}{c_2}\right)^2}, \frac{-d_1 - c_2 d_2}{c_2}\right) + \arctan2(-p_x, p_y)$$

$$\theta_6 = \arctan2\left(\frac{1}{s_2} (c_1 r_{2,2} - s_1 r_{1,2}), \frac{1}{s_2} (s_1 r_{1,1} - c_1 r_{2,1})\right)$$

After eliminating θ_2 we have:

$$\theta_2 = \arctan2(k_1 k_3 - k_1 k_4, k_1 k_3 + k_1 k_4)$$

$$\theta_3 = \arctan2\left(\pm \sqrt{1 - \frac{1}{\alpha_1^2} (k_1^2 + k_2^2 - k_3^2 - k_4^2 - \alpha_1^2)}, \frac{1}{\alpha_1} (k_1^2 + k_2^2 - k_3^2 - k_4^2 - \alpha_1^2)\right) + \arctan2(-k_6, k_5)$$

For this motion planning problem multiple choices are made:

- The first and most important one is the usage of joint space.

This space prevents the appearance of singularities and so highly simplifying the problem.

It also drastically reduce the calculation time as inverse kinematic in our case is time consuming.

While reducing the computation of inverse kinematics it also imply reducing the amount of choices

needed in case of multiple solutions.

- The second choice is the method of selection in case of multiple solutions of the inverse-kinematic.

To make this choice let's define : $C(\theta, \theta') = \sum_{i=1}^n (x_i - x'_i)^2$

Here the idea is to use C as a cost function to estimate the cost of a movement from θ to θ' .

And θ, θ' being vectors of joint angles. The cost here is a sort of weighted distance making a movement

of the base more important than one of the wrist.

We then obviously choose the movement with the lowest cost.

In our problem we will choose the cheapest combination of movement from A to B and then C.

- To make sure that the path stays in the workspace we have to make sure that the joint angles

do not go out of band. To address this problem we will duplicate the inverse kinematic solutions using modulus

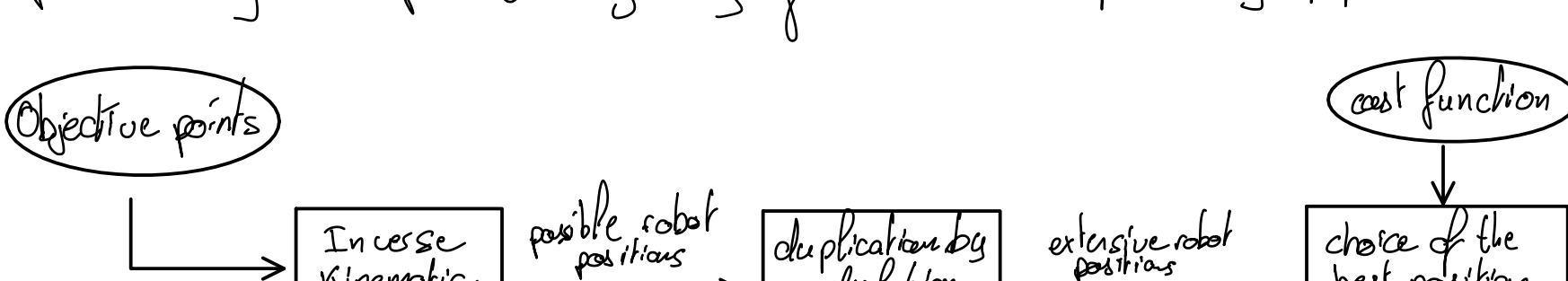
while staying in the defined boundaries (if we have $\theta_i = 2\pi$ we duplicate the solution to $\theta_i = 140^\circ$). After this step

we will consider the joint angles as not modifiable values. This permit us to have all possible modulus

values for joint angles without the risk of going out of range as going from $\theta_i = 2\pi$ to $\theta_i = 140^\circ$ will be

considered a 360° movement.

Following this plan the beginning of the motion planning pipeline looks something like this:



Next is the actual motion planning.

We know $t_{acc} = 0.5 \text{ sec}$: half transition time

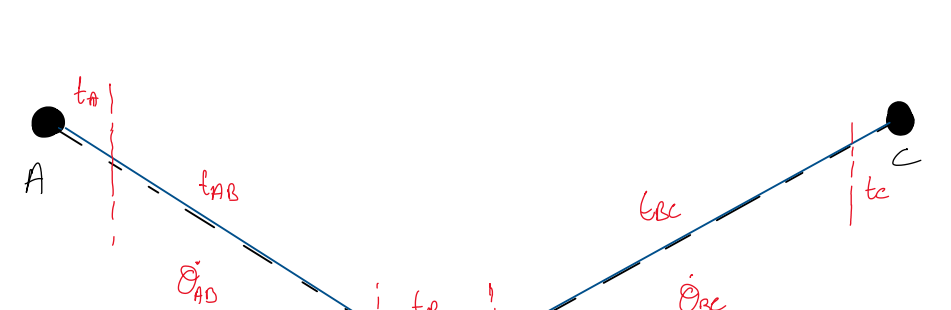
$t_{AB} = 5.0 \text{ sec}$: overall duration from A to B

$t_{BC} = 4.0 \text{ sec}$: overall duration from B to C

for the math computation we will use a sampling time of 0.01 sec .

Let's call $(\theta_A, \theta_B, \theta_C) \in \mathbb{R}^6$ the chosen positions for each point.

The problem is now to find the red unknown using the values defined just before.



First we know the half transition time t_{acc} , so we know $t_A = t_C = t_{acc}$ and $t_B = 2t_{acc}$

Furthermore $t_{AB} = t_{AB} - t_A - \frac{1}{2}t_B = t_{AB} - 2t_{acc}$

$t_{BC} = t_{BC} - \frac{1}{2}t_B - t_C = t_{BC} - 2t_{acc}$

Hence we get $\theta_B = \frac{\theta_C - \theta_A}{t_{BC} - \frac{1}{2}t_B} = \frac{\theta_C - \theta_A}{t_{BC} - 2t_{acc}}$

$\theta_{AB} = \frac{\theta_B - \theta_A}{t_{AB} - \frac{1}{2}t_B}$

Now we need the acceleration during the different transition.

We will consider the accelerations constant.

$\ddot{\theta}_A = 0$ and $\ddot{\theta}$ is continuous.

So $\ddot{\theta}_B = \ddot{\theta}(0) = 0$, $\ddot{\theta}_{AB} = \ddot{\theta}(t_A)$

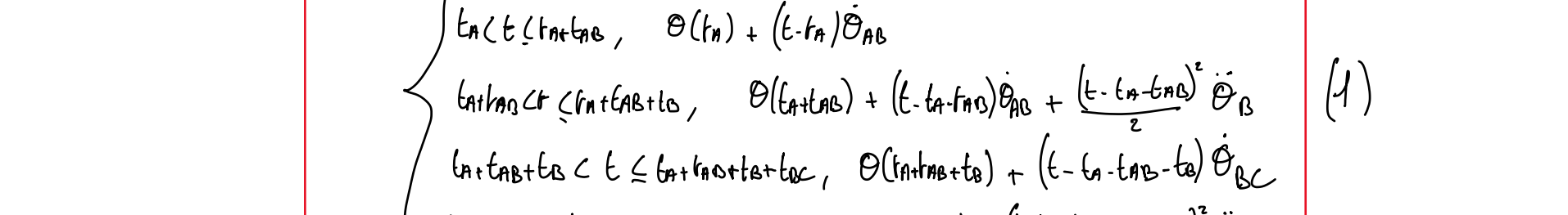
And $\ddot{\theta}_B = \frac{\ddot{\theta}_{AB}}{t_A}$

Using the same logic $\ddot{\theta}_C = \frac{\ddot{\theta}_{BC}}{t_C}$

And $\ddot{\theta}_B = \frac{\ddot{\theta}_{BC} - \ddot{\theta}_{AB}}{2t_{acc}}$

Finally we get $\theta(t) = \begin{cases} t \leq t_A, t_A t_{AB} + \frac{1}{2} \ddot{\theta}_A \\ t_A < t < t_A + t_{AB}, \quad \theta(t_A) + (t - t_A) \ddot{\theta}_{AB} \\ t_A + t_{AB} < t < t_A + t_{AB} + t_{BC}, \quad \theta(t_A + t_{AB}) + (t - t_A - t_{AB}) \ddot{\theta}_B \\ t_A + t_{AB} + t_{BC} < t < t_A + t_{AB} + t_{BC} + t_{BC}, \quad \theta(t_A + t_{AB} + t_{BC}) + (t - t_A - t_{AB} - t_{BC}) \ddot{\theta}_C \\ t_A + t_{AB} + t_{BC} + t_{BC} < t, \quad \theta(t_A + t_{AB} + t_{BC} + t_{BC}) + \frac{(t - t_A - t_{AB} - t_{BC} - t_{BC})^2}{2} \ddot{\theta}_C \end{cases}$

Finally the complete pipeline is:



Results:

